

Civic.ai RAG Birth and Death Registration Domain

A grounded Bangla-first Retrieval-Augmented Generation system inspired by the NextRAG paper, adapted for Bangladeshi government-service information.

187

indexed evidence chunks

1

active domain: birth/death
registration

3

local Ollama LLMs available for
comparison

Repository:

Shihabsarker93/civic_ai_rag

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8d83e2a Add correction fee supplemental
evidence

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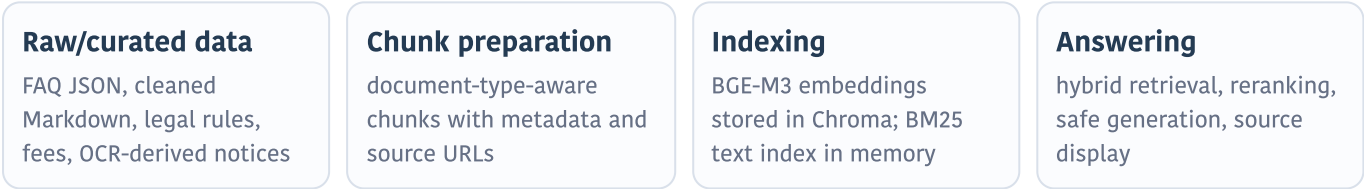
1. Executive Summary

We now have a working local Civic.ai RAG prototype for Bangladeshi birth/death registration. The current system is not a generic chatbot: it retrieves indexed government-service evidence, reranks it, and answers with source IDs and official/source links. For high-risk civic questions such as fees, correction, required documents, deadlines, and lost certificates, CivicRAG uses controlled evidence-based answer paths to reduce hallucination.

The project direction has been narrowed from a multi-domain prototype to a clean single active domain first: birth/death registration. Previous passport and BRTA data were removed from the active pipeline so that the current vector database is clean and domain-specific.

The implementation is closest to the NextRAG paper at the architectural level: simple baseline versus proposed advanced pipeline, multiple LLM backends, bilingual/cross-lingual goals, and grounded retrieval. However, the system is not yet at the paper's evaluation maturity because a labeled gold QA set and numeric metrics are still pending.

2. Current System Architecture



Module	Current implementation	Why it matters
Domain config	domains/birth_death_registration/config.json	Keeps each future government-service domain separate without creating separate repositories.
Chunk preparation	domains/birth_death_registration/scripts/prepare_chunks.py	Builds retrieval-friendly evidence units based on document type.
Embedding	BAAI/bge-m3, normalized, CPU, local files only	Strong multilingual embedding candidate suitable for Bangla-English retrieval experiments.
Vector database	Chroma collection birth_death_registration_chunks	Persistent local semantic search without external API keys.
Lexical retrieval	BM25-style sparse retrieval inside HybridRetriever	Protects exact Bangla keywords such as fees, sections, rules, and official terms.
Fusion	Dense + BM25 with RRF, weights 0.55 dense / 0.45 BM25	Balances semantic matching with exact keyword matching.
Reranking	BAAI/bge-reranker-v2-m3 plus civic intent boosts	Moves application-process, correction, fee, and legal chunks above weak FAQ matches when needed.
Generation	Ollama local models: llama3.2, llama3, qwen2.5:7b	Allows offline thesis experiments without API keys.
Web UI	Local app at http://127.0.0.1:7860	Supports model and pipeline switching, answer display, source IDs, and official/source links.

3. Data and Index Status

The current active knowledge base contains curated FAQ JSON, cleaned Markdown, legal documents, fee tables, application/correction guides, portal guidance, and OCR-derived sources. The Chroma vector database count matches the chunk JSONL count: **187 records**.

Document type	Chunks	Role in retrieval
legal_rules	45	Rules, fees, official procedures, duplicate/copy certificates
legal_act	30	Core legal authority such as registration deadlines and correction authority
application_process	30	Step-by-step online birth registration process and required information
faq	26	Citizen-facing questions such as manual-to-online migration and data mismatch
correction_process	12	Name, parent information, address, and other correction guidance
guidelines_ocr	12	OCR-derived guideline source; useful but lower confidence
portal_summary	9	Cleaned portal/service/support summaries
general_guidance	8	Verification and practical guidance pages
fee_row	7	Exact row-level fee answers, e.g. 100 taka and 50 taka correction fees
correction_notice_ocr	6	OCR/manual-cleaned correction notice for special BDRIS correction features
fees_table + faq_preamble	2	Full fee table and FAQ context

Source files currently represented

Main indexed sources include: `birth_and_death_registration_rules_2018.md`, `birth_and_death_registration_act_2004.md`, `birth_registration_application_process_cleaned.md`, `faqs_birth_death_registration_01.json`, `faqs_birth_death_registration_02.json`, `application_for_birth_information_correction_cleaned.md`, `birth_and_death_registration_fees_cleaned.md`, `know_this_01_cleaned.md`, `birth_and_death_registration_guidelines_2021_ocr.md`, and `notice_birth_and_death_registration_certificate_correction_steps_ocr.md`.

Important: each chunk separates **retrieval_text** from **answer evidence**. Search aliases are used for retrieval only, while the LLM receives original Bangla evidence content. This directly addresses a hallucination risk we observed earlier.

4. Chunking Strategy

We chose document-type-aware chunking instead of one universal chunk size. This aligns better with government-service data, where FAQs, laws, fee tables, notices, and process guides have different retrieval behavior.

Data type	Chunking choice	Reason
FAQ JSON	One question-answer pair per chunk	Keeps user intent and answer together; strong for exact citizen questions.
Legal act/rules	One ধারা or বিধি per chunk, split if long	Preserves legal authority and section-level citation.
Fee table	Full table chunk + row-level fee chunks	Supports broad fee questions and exact amount questions.
Application process	Heading/step-based chunks	Improves “কীভাবে করব” procedural retrieval.
Correction process	Scenario/section chunks	Separates parent-name, address, birth-date, and contact scenarios.
Portal/homepage content	Only citizen-service/support sections retained	Removes menus, footers, news lists, and portal noise.
Scanned PDF/OCR	OCR to Markdown, then structured chunks	Allows scanned notices/guidelines to be indexed, but with lower confidence metadata.

This is thesis-defensible because it allows ablation: FAQ-only vs legal-only vs process-guide chunks, OCR on/off, row-level fee chunks on/off, and simple fixed-size chunking vs structured document-aware chunking.

5. What Makes CivicRAG “Ours”

The LLMs are not our model. Our contribution is the RAG system: data pipeline, chunking strategy, multilingual retrieval setup, reranking, safe answer generation, citations, and evaluation plan.

Simple RAG baseline

- Dense vector search only.
- Less domain-aware reranking.
- LLM has more freedom to synthesize.
- Useful as a tutorial/baseline comparison.

CivicRAG / proposed pipeline

- Hybrid dense + BM25 retrieval.
- RRF fusion and BGE reranking.
- Civic intent boosts for fees, correction, documents, lost certificate, and online visibility.
- Safe evidence-based answer paths for high-risk civic questions.

If all three models give the same answer in CivicRAG, that is not automatically a problem. For high-risk fee/procedure questions, the system intentionally uses controlled grounded answers instead of letting each local LLM freely rewrite the response. This improves factual stability.

6. Current LLM and Prompt Settings

Model	Role in thesis	Current status
llama3.2	Fast local baseline	Available through Ollama; default generator label.
llama3	Stronger local LLaMA baseline	Available for comparison.
qwen2.5:7b	Strong multilingual alternative	Available for comparison; previously showed language drift when unconstrained.

Generation settings are conservative: temperature **0.05**, top-p **0.9**, max prediction **1200**, repeat penalty **1.18**. Low temperature supports consistency and factual grounding.

7. Manual Sanity Test Results

We tested eight Bangla questions representing realistic citizen-service behavior. These are not final numeric evaluation results; they are manual sanity checks to see whether retrieval and answers are moving in the right direction.

Question	Observed result	Judgment
এতিম শিশুর জন্ম নিবন্ধনে ফি লাগবে কি?	Correctly says fee may be fully/partially waived under rule evidence; not “always free”.	Good
জন্ম তারিখ ভুল হলে কীভাবে সংশোধন করব?	Now includes correction application, BDRIS feature, ১০০ টাকা fee, and warning that full steps are not in the dataset.	Good
জন্মের কত দিনের মধ্যে নিবন্ধন করতে হবে?	Correctly answers ৪৫ দিনের মধ্যে with act/rule evidence.	Good
জন্ম নিবন্ধনের জন্য কী কী কাগজপত্র লাগে?	Uses application-process and rule evidence for address/proof documents.	Good
জন্ম নিবন্ধন সনদ হারিয়ে গেলে কী করতে হবে?	Retrieves সনদের প্রতিলিপি rule and mentions ৭ কার্য দিবস.	Good
অনলাইনে জন্ম নিবন্ধন আবেদন কীভাবে করতে হয়?	Gives step-based online application answer and includes the BDRIS link.	Good
অনলাইন করা আছে কিন্তু অনলাইনে দেখাচ্ছেনা কেন?	Combines verification guidance, manual-to-online migration, and data mismatch explanation.	Mostly good
নাম ঠিকানা সব ভুল, কীভাবে ঠিক করবে?	Uses correction-process evidence and includes ৫০ টাকা correction fee with source.	Good

Current weakness: these checks are not yet formal accuracy scores. The next stage must create a labeled QA set and run retrieval/answer metrics.

8. How Close We Are to the NextRAG Paper

NextRAG-like component	Current Civic.ai status	Gap
Vanilla/simple RAG baseline	Implemented through Simple RAG option.	Needs formal metric comparison.
Proposed advanced RAG method	CivicRAG hybrid retrieval + reranking + safer answers.	Needs formal naming and method diagram for thesis.
Multiple LLM comparison	llama3.2, llama3, qwen2.5:7b available locally.	Need batch comparison script outputs for all test questions.
Bilingual/cross-lingual goal	Config supports bn/en/mixed; BGE-M3 is multilingual.	English and code-mixed test set still needed.
Domain-specific QA categories	Implicit categories exist: fees, correction, application, legal, FAQ, lost certificate.	Need explicit evaluation categories like factual, procedural, fee, correction, out-of-context.
Numeric evaluation table	Not completed yet.	Need Recall@k, MRR, nDCG, answer faithfulness, and semantic similarity/BERTScore-style metrics.

9. Recommended Next Steps

1. **Create a gold QA dataset.** Start with 50 to 100 Bangla questions across fee, application, correction, legal, FAQ, online visibility, lost certificate, and out-of-context categories.
2. **Label expected source IDs.** For retrieval metrics, each question needs one or more relevant source chunk IDs.
3. **Run retrieval metrics.** Measure Recall@3, Recall@5, MRR, and nDCG for Simple RAG vs CivicRAG.
4. **Run answer metrics.** Use reference answers plus qualitative faithfulness checks; BERTScore or semantic similarity can be supportive, not the only evidence.
5. **Add English and code-mixed tests.** This is necessary because the thesis goal is bilingual/cross-lingual Civic.ai, not Bangla-only service search.
6. **Do ablations.** Compare dense only, BM25 only, hybrid without reranker, hybrid with reranker, and CivicRAG with safe answer paths.
7. **Review OCR guideline chunks.** Keep OCR data marked as lower confidence until manually verified.
8. **Prepare a method diagram.** Show the separation between ingestion, cleaning, chunking, embedding, retrieval, reranking, generation, and evaluation.

10. Current Repository Pointers

Purpose	Path
Active domain config	domains/birth_death_registration/config.json
Chunk preparation script	domains/birth_death_registration/scripts/prepare_chunks.py
OCR conversion script	domains/birth_death_registration/scripts/ocr_scanned_pdfs.py
Prepared chunks	domains/birth_death_registration/data/interim/birth_death_chunks.jsonl
Vector database	domains/birth_death_registration/data/processed/chroma_birth_death_bge_m3
Main pipeline	src/pipeline.py
Hybrid retrieval	src/retrieval/hybrid_retriever.py
Reranking	src/reranking/reranker.py
Local web app	app.py
Raw data audit	docs/birth_death_raw_data_audit.md

Bottom line: Civic.ai is now a working, locally runnable, source-grounded Bangla RAG prototype for birth/death registration. The next big academic milestone is not adding more tricks; it is building a clean evaluation set and producing defensible comparison tables.